STACKS, DISCRIMINANTS: LEMMA-DOMINATE-FACTORIZATIONS

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 320-329 of sources/stacks-discriminant/pinned-original-source.tex; proof span is lines 331-354, the complete author proof (22 body lines).

Editorial note. Navigation stubs: 1 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 31 of queue_discriminant.json (pb=876, stmt 320-329, proof 331-354).

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 02JJ: sources/stacks-discriminant/pinned-original-source.tex, source lines 354–354. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let $A \to B$ be a quasi-finite ring map. Given two factorizations $A \to B' \to B$ and $A \to B'' \to B$ with $A \to B'$ and $A \to B''$ finite and $\mathrm{Spec}(B) \to \mathrm{Spec}(B')$ and $\mathrm{Spec}(B) \to \mathrm{Spec}(B'')$ open immersions, there exists an A-subalgebra $B''' \subset B$ finite over A such that $\mathrm{Spec}(B) \to \mathrm{Spec}(B''')$ an open immersion and $B' \to B$ and $B'' \to B$ factor through B'''.*

Proof. Let $B''' \subset B$ be the A-subalgebra generated by the images of $B' \to B$ and $B'' \to B$. As B' and B'' are each generated by finitely many elements integral over A, we see that B''' is generated by finitely many elements integral over A and we conclude that B''' is finite over A (Algebra, Lemma 02JJ). Consider the maps

$$B = B' \otimes_{B'} B \to B''' \otimes_{B'} B \to B \otimes_{B'} B = B$$

The final equality holds because $\mathrm{Spec}(B) \to \mathrm{Spec}(B')$ is an open immersion (and hence a monomorphism). The second arrow is injective as $B' \to B$ is flat. Hence both arrows are isomorphisms. This means that

$$\begin{array}{ccc} \mathrm{Spec}(B''') & \longleftarrow & \mathrm{Spec}(B) \\ \downarrow & & \downarrow \\ \mathrm{Spec}(B') & \longleftarrow & \mathrm{Spec}(B) \end{array}$$

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is cartesian. Since the base change of an open immersion is an open immersion we conclude. □